

# Package ‘manMetaVAR’

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Coefficients

**Version** 0.9.7

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Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.  
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| Check | <i>Check Replication</i> |
|-------|--------------------------|

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**Description**

Check Replication

**Usage**

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

**Arguments**

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                           |
| repid         | Positive integer. Replication ID.                                    |
| output_folder | Character string. Output folder.                                     |
| naive         | Logical. If naive = TRUE, fit the naive approach.                    |
| metavar       | Logical. If metavar = TRUE, fit the model using the metaDyn package. |
| mplus         | Logical. If mplus = TRUE, fit the model using DSEM in Mplus.         |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

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|               |                                     |
|---------------|-------------------------------------|
| CheckFitDTVAR | <i>Check Replication - FitDTVAR</i> |
|---------------|-------------------------------------|

---

**Description**

Check Replication - FitDTVAR

**Usage**

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

**Arguments**

|               |                                                     |
|---------------|-----------------------------------------------------|
| taskid        | Positive integer. Task ID.                          |
| repid         | Positive integer. Replication ID.                   |
| output_folder | Character string. Output folder.                    |
| suffix        | Character string. Output of manCTMed:::SimSuffix(). |

**Details**

This function is executed via the Check function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

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|                 |                                       |
|-----------------|---------------------------------------|
| CheckFitMetaVAR | <i>Check Replication - FitMetaVAR</i> |
|-----------------|---------------------------------------|

---

**Description**

Check Replication - FitMetaVAR

**Usage**

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

**Arguments**

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                        |
| repid         | Positive integer. Replication ID.                                 |
| output_folder | Character string. Output folder.                                  |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> . |

**Details**

This function is executed via the Check function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|               |                                     |
|---------------|-------------------------------------|
| CheckFitMplus | <i>Check Replication - FitMplus</i> |
|---------------|-------------------------------------|

---

**Description**

Check Replication - FitMplus

**Usage**

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

**Arguments**

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                        |
| repid         | Positive integer. Replication ID.                                 |
| output_folder | Character string. Output folder.                                  |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> . |

**Details**

This function is executed via the Check function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|               |                                     |
|---------------|-------------------------------------|
| CheckFitNaive | <i>Check Replication - FitNaive</i> |
|---------------|-------------------------------------|

---

**Description**

Check Replication - FitNaive

**Usage**

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

**Arguments**

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                       |
| repid         | Positive integer. Replication ID.                                |
| output_folder | Character string. Output folder.                                 |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> . |

**Details**

This function is executed via the Check function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Compress

*Compress Replication*

---

**Description**

Compress Replication

**Usage**

```
Compress(taskid, repid, output_folder)
```

**Arguments**

|               |                                   |
|---------------|-----------------------------------|
| taskid        | Positive integer. Task ID.        |
| repid         | Positive integer. Replication ID. |
| output_folder | Character string. Output folder.  |

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

FigBias

*Plot Relative Bias***Description**

Plot relative bias for the fixed and random effects.

**Usage**

```
FigBias(
  results,
  bias = FALSE,
  method = c("BMLVAR", "MetaVAR", "Naive"),
  parameters = "both",
  ylim = c(-0.15, 0.15),
  x_lab_size = 6
)
```

**Arguments**

|            |                                                                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| results    | results data frame.                                                                                                                                                                |
| bias       | Logical. If bias = TRUE, plot absolute bias. If bias = FALSE, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias. |
| method     | Character vector. Methods to include in the plot.                                                                                                                                  |
| parameters | Character string. Parameters to plot. Valid values include "fixed", "random", and "both".                                                                                          |
| ylim       | Numeric vector of length 2. Lower and upper limits for the y-axis.                                                                                                                 |
| x_lab_size | Numeric. x-axis label size.                                                                                                                                                        |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

**Examples**

```
## Not run:
data(results, package = "manMetaVAR")
FigBias(results)

## End(Not run)
```

---

FigCoverage

*Plot Coverage Probabilities*


---

## Description

Plot coverage probabilities for the fixed and random effects.

## Usage

```
FigCoverage(
  results,
  method = c("BMLVAR", "MetaVAR", "Naive"),
  parameters = "both",
  ylim = c(0.5, 1),
  x_lab_size = 6
)
```

## Arguments

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| results    | results data frame.                                                                       |
| method     | Character vector. Methods to include in the plot.                                         |
| parameters | Character string. Parameters to plot. Valid values include "fixed", "random", and "both". |
| ylim       | Numeric vector of length 2. Lower and upper limits for the y-axis.                        |
| x_lab_size | Numeric. x-axis label size.                                                               |

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Figure Functions: [FigBias\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

## Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigCoverage(results)

## End(Not run)
```



**Description**

Plot statistical power probabilities for the fixed and random effects.

**Usage**

```
FigPower(  
  results,  
  method = c("BMLVAR", "MetaVAR", "Naive"),  
  parameters = "both",  
  ylim = c(0, 1),  
  x_lab_size = 7.5  
)
```

**Arguments**

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| results    | results data frame.                                                                       |
| method     | Character vector. Methods to include in the plot.                                         |
| parameters | Character string. Parameters to plot. Valid values include "fixed", "random", and "both". |
| ylim       | Numeric vector of length 2. Lower and upper limits for the y-axis.                        |
| x_lab_size | Numeric. x-axis label size.                                                               |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigRMSE\(\)](#)

**Examples**

```
## Not run:  
data(results, package = "manMetaVAR")  
FigPower(results)  
  
## End(Not run)
```

FigRMSE

*Plot RMSE***Description**

Plot RMSE for the fixed and random effects.

**Usage**

```
FigRMSE(
  results,
  method = c("BMLVAR", "MetaVAR", "Naive"),
  parameters = "both",
  ylim = c(0, 0.25),
  x_lab_size = 6
)
```

**Arguments**

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| results    | results data frame.                                                                       |
| method     | Character vector. Methods to include in the plot.                                         |
| parameters | Character string. Parameters to plot. Valid values include "fixed", "random", and "both". |
| ylim       | Numeric vector of length 2. Lower and upper limits for the y-axis.                        |
| x_lab_size | Numeric. x-axis label size.                                                               |

**Author(s)**

Ivan Jacob Agaloos Pesigan

**See Also**

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#)

**Examples**

```
## Not run:
data(results, package = "manMetaVAR")
FigRMSE(results)

## End(Not run)
```

FitDTVAR

*Fit the Model using the fitVARMxID Package***Description**

The function fits the model using the [fitVARMxID::fitVARMxID](#) package.

**Usage**

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

**Arguments**

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |
| seed   | Integer. Random seed.                                       |

**See Also**

Other Model Fitting Functions: [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR

*Multivariate Meta-Analysis using the metaDyn Package***Description**

The function performs multivariate meta-analysis using the [metaDyn::metaDyn](#) package.

**Usage**

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

**Arguments**

|        |                                                              |
|--------|--------------------------------------------------------------|
| fit    | R object. Output of the <a href="#">FitDTVAR()</a> function. |
| ncores | Positive integer. Number of cores to use.                    |
| seed   | Integer. Random seed.                                        |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

---

FitMplus

---

*Fit the Model using Mplus*


---

**Description**

The function fits the model using Mplus.

**Usage**

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

**Arguments**

|           |                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------|
| data      | R object. Output of the <a href="#">GenData()</a> function.                                                        |
| chains    | Integer. Number of chains.                                                                                         |
| iter      | Integer. Number of iterations.                                                                                     |
| fscores   | Integer. Number of iterations for factor scores.                                                                   |
| plot      | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                                              |
| wd        | Character string. Working directory.                                                                               |
| mplus_bin | Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary. |
| ncores    | Positive integer. Number of cores to use.                                                                          |
| seed      | Integer. Random seed.                                                                                              |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

---

FitNaive

*Naive*


---

**Description**

The function performs the naive “fit-many-then-summarize”.

**Usage**

```
FitNaive(fit)
```

**Arguments**

|     |                                                              |
|-----|--------------------------------------------------------------|
| fit | R object. Output of the <a href="#">FitDTVAR()</a> function. |
|-----|--------------------------------------------------------------|

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitNaive(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

---

GenData

*Simulate Data*


---

**Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

**Usage**

```
GenData(taskid, seed = NULL)
```

**Arguments**

|        |                            |
|--------|----------------------------|
| taskid | Positive integer. Task ID. |
| seed   | Integer. Random seed.      |

**Examples**

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

---

`manmetavar-data-methods`*Methods for Objects of Class `manmetavar.data`*

---

**Description**

This page documents the available methods for objects of class `manmetavar.data`.

**Usage**

```
## S3 method for class 'manmetavar.data'
print(x, ...)
```

```
## S3 method for class 'manmetavar.data'
summary(object, ...)
```

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

**Arguments**

|                     |                                                |
|---------------------|------------------------------------------------|
| <code>x</code>      | Object of class <code>manmetavar.data</code> . |
| <code>...</code>    | additional arguments.                          |
| <code>object</code> | Object of class <code>manmetavar.data</code> . |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

`manmetavar-dtvar-methods`*Methods for Objects of Class `manmetavar.dtvar`*

---

**Description**

This page documents the available methods for objects of class `manmetavar.dtvar`.

**Usage**

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
```

```
    nu = TRUE,
    psi = TRUE,
    theta = TRUE,
    ncores = NULL,
    ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
```



```
    ...  
  )
```

### Arguments

|        |                                                                                                                                                                           |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.dtvvar</code> .                                                                                                                          |
| mu     | Logical. If <code>mu = TRUE</code> , include estimates of the mu vector, if available. If <code>mu = FALSE</code> , exclude estimates of the mu vector.                   |
| alpha  | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.       |
| beta   | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.           |
| nu     | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.                   |
| psi    | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.               |
| theta  | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.       |
| ncores | Positive integer. Number of cores to use.                                                                                                                                 |
| ...    | additional arguments.                                                                                                                                                     |
| robust | Logical. If <code>TRUE</code> , use robust (sandwich) sampling variance-covariance matrix. If <code>FALSE</code> , use normal theory sampling variance-covariance matrix. |
| x      | Object of class <code>manmetavar.dtvvar</code> .                                                                                                                          |
| means  | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                                      |
| digits | Integer indicating the number of decimal places to display.                                                                                                               |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods

*Methods for Objects of Class `manmetavar.metavar`*

---

### Description

This page documents the available methods for objects of class `manmetavar.metavar`.

This page documents the available methods for objects of class `manmetavar.naive`.

**Usage**

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

**Arguments**

|        |                                                                                                                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.naive</code> .                                                                                                                                                                                |
| ...    | additional arguments.                                                                                                                                                                                                          |
| robust | Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available. |
| x      | Object of class <code>manmetavar.naive</code> .                                                                                                                                                                                |
| alpha  | Numeric vector. Significance level $\alpha$ .                                                                                                                                                                                  |
| digits | Integer indicating the number of decimal places to display.                                                                                                                                                                    |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.                                                          |
| level  | the confidence level required.                                                                                                                                                                                                 |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

manmetavar-mplus-methods

*Methods for Objects of Class manmetavar.mplus*


---

## Description

This page documents the available methods for objects of class `manmetavar.mplus`.

## Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

## Arguments

|                     |                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class <code>manmetavar.mplus</code> .                                                                                                                       |
| <code>median</code> | Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.                               |
| <code>burnin</code> | Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.                                                               |
| <code>...</code>    | additional arguments.                                                                                                                                                 |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ .                                                                                                                         |
| <code>digits</code> | Integer indicating the number of decimal places to display.                                                                                                           |
| <code>parm</code>   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |

|            |                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| level      | the confidence level required.                                                                                                                   |
| x          | Object of class <code>manmetavar.mplus</code> .                                                                                                  |
| what       | Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots. |
| legend_loc | Character string. Legend location.                                                                                                               |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|       |                         |
|-------|-------------------------|
| model | <i>Model Parameters</i> |
|-------|-------------------------|

---

**Description**

Model Parameters

**Usage**

```
data(model)
```

**Format**

A list with 15 elements:

**k** Number of variables.

**mu\_mu** Mean of the set-point vector  $\mu$ .

**mu\_sigma** Covariance matrix of the parameter  $\mu$ .

**mu\_sigma\_l** Cholesky factor of the covariance matrix of the parameter  $\mu$ .

**beta\_mu** Mean of lagged coefficients matrix  $\beta$ .

**beta\_sigma** Covariance matrix of the parameter  $\text{vec}(\beta)$ .

**beta\_sigma\_l** Cholesky factor of the covariance matrix of the parameter  $\text{vec}(\beta)$ .

**psi** Process noise covariance matrix  $\Psi$ .

**psi\_l** Cholesky factor of the process noise covariance matrix  $\Psi$ .

**psi\_d\_ldl** uc\_d of the LDL' decomposition of the process noise covariance matrix  $\Psi$ . See [fitVARMxID::LDL\(\)](#).

**psi\_l\_ldl** s\_l of the LDL' decomposition of the process noise covariance matrix  $\Psi$ . See [fitVARMxID::LDL\(\)](#).

**ma\_fixed** Vector of fixed effects  $\theta = [\mu, \text{vec}(\beta)]'$

**ma\_random** Matrix of random effects  $\theta = [\mu, \text{vec}(\beta)]'$

**ma\_random\_d\_ldl** uc\_d of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

**ma\_random\_l\_ldl** s\_l of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|            |                         |
|------------|-------------------------|
| MplusInput | Create Mplus Input File |
|------------|-------------------------|

---

**Description**

The function creates an Mplus input file.

**Usage**

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

**Arguments**

|                 |                                                                       |
|-----------------|-----------------------------------------------------------------------|
| fn_data         | Character string. Filename for data file.                             |
| fn_posterior    | Character string. Filename for posterior output.                      |
| fn_factorscores | Character string. Filename for factor scores output.                  |
| chains          | Integer. Number of chains.                                            |
| iter            | Integer. Number of iterations.                                        |
| fscores         | Integer. Number of iterations for factor scores.                      |
| plot            | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file. |
| ncores          | Positive integer. Number of cores to use.                             |
| seed            | Integer. Random seed.                                                 |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#)

**Examples**

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
```

```
      chains = 2L,  
      iter = 40000L,  
      fscores = 1000L,  
      plot = TRUE,  
      ncores = NULL,  
      seed = 42  
    )  
  )
```

---

|        |                              |
|--------|------------------------------|
| params | <i>Simulation Parameters</i> |
|--------|------------------------------|

---

**Description**

Simulation Parameters

**Usage**

data(params)

**Format**

A dataframe with 12 rows and 3 columns:

- taskid** Simulation Task ID.
- n** Sample size.
- time** Number of measurement occasions. If NA, measurement occasions vary across IDs.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|         |                           |
|---------|---------------------------|
| results | <i>Simulation Results</i> |
|---------|---------------------------|

---

**Description**

Simulation Results

**Usage**

data(results)

**Format**

A dataframe with I rows and J columns:

**taskid** Simulation Task ID.

**replications** Number of replications.

**parnames** Parameter names.

**parameter** Population parameter value.

**method** Method used to fit the data.

**n** Sample size.

**time** Number of measurement occasions.

**ci** Confidence/credible intervals method.

**est** Mean parameter estimate.

**se** Mean standard error.

**t** Mean test statistic.

**p** Mean  $p$ -value.

**ll** Mean lower limit of the 95% confidence/credible interval.

**ul** Mean upper limit of the 95% confidence/credible interval.

**sig** Proportion of statistically significant results.

**theta\_hit** Proportion of replications where the confidence intervals contained the population parameter.

**sq\_error** Mean squared error.

**bias** Bias.

**rel\_bias** Relative bias.

**se\_bias** Bias in standard error estimate.

**rel\_se\_bias** Relative bias in standard error estimate.

**rmse** Root mean square error.

**coverage** Coverage probability.

**power** Statistical power.

**par\_idx** Parameter index.

**Author(s)**

Ivan Jacob Agaloos Pesigan

Sim

*Simulation Replication***Description**

Simulation Replication

**Usage**

```

Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)

```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| repid         | Positive integer. Replication ID.                                                         |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed          | Integer. Random seed.                                                                     |
| data          | Logical. Simulate data.                                                                   |
| naive         | Logical. If naive = TRUE, fit the naive approach.                                         |
| metavar       | Logical. If metavar = TRUE, fit the model using the metaDyn package.                      |
| mplus         | Logical. If mplus = TRUE, fit the model using DSEM in Mplus.                              |
| chains        | Integer. Number of chains.                                                                |
| iter          | Integer. Number of iterations.                                                            |
| fscores       | Integer. Number of iterations for factor scores.                                          |
| plot          | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                     |



**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitDTVAR

*Simulation Replication - FitDTVAR*

---

**Description**

Simulation Replication - FitDTVAR

**Usage**

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in output_folder.                                                                  |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|               |                                            |
|---------------|--------------------------------------------|
| SimFitMetaVAR | <i>Simulation Replication - FitMetaVAR</i> |
|---------------|--------------------------------------------|

---

**Description**

Simulation Replication - FitMetaVAR

**Usage**

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

`SimFitMplus`*Simulation Replication - FitMplus*

---

**Description**

Simulation Replication - FitMplus

**Usage**

```
SimFitMplus(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  chains,  
  iter,  
  fscores,  
  plot  
)
```

**Arguments**

|                            |                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>taskid</code>        | Positive integer. Task ID.                                                                                            |
| <code>repid</code>         | Positive integer. Replication ID.                                                                                     |
| <code>output_folder</code> | Character string. Output folder.                                                                                      |
| <code>seed</code>          | Integer. Random seed.                                                                                                 |
| <code>suffix</code>        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| <code>overwrite</code>     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| <code>integrity</code>     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |
| <code>chains</code>        | Integer. Number of chains.                                                                                            |
| <code>iter</code>          | Integer. Number of iterations.                                                                                        |
| <code>fscores</code>       | Integer. Number of iterations for factor scores.                                                                      |
| <code>plot</code>          | Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.                      |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitNaive

*Simulation Replication - FitNaive*

---

**Description**

Simulation Replication - FitNaive

**Usage**

```
SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|       |                             |
|-------|-----------------------------|
| SimFN | <i>Simulation File Name</i> |
|-------|-----------------------------|

---

**Description**

Simulation File Name

**Usage**

```
SimFN(output_type, output_folder, suffix)
```

**Arguments**

|               |                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------|
| output_type   | Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-mplus". |
| output_folder | Character string. Output folder.                                                                                 |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> .                                                 |

**Value**

Returns a character string file name with the `output_folder` in the OS-specific format.

---

|            |                                         |
|------------|-----------------------------------------|
| SimGenData | <i>Simulation Replication - GenData</i> |
|------------|-----------------------------------------|

---

**Description**

Simulation Replication - GenData

**Usage**

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed::SimSuffix()</code> .                                                      |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|         |                                |
|---------|--------------------------------|
| SimProj | <i>Simulation Project Name</i> |
|---------|--------------------------------|

---

**Description**

Simulation Project Name

**Usage**

SimProj()

**Value**

Returns the project name as a character string.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |                |
|-----|----------------|
| Sum | <i>Summary</i> |
|-----|----------------|

---

**Description**

Summary

**Usage**

```
Sum(  
  taskid,  
  reps,  
  output_folder,  
  overwrite,  
  integrity,  
  naive,  
  metavar_normal,  
  metavar_robust,  
  mplus,  
  ncores  
)
```

**Arguments**

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| taskid         | Positive integer. Task ID.                                                                               |
| reps           | Positive integer. Number of replications.                                                                |
| output_folder  | Character string. Output folder.                                                                         |
| overwrite      | Logical. Overwrite existing output in output_folder.                                                     |
| integrity      | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.                |
| naive          | Logical. If naive = TRUE, summarize naive estimates.                                                     |
| metavar_normal | Logical. If metavar_normal = TRUE, summarize metaVAR model using normal confidence intervals.            |
| metavar_robust | Logical. If metavar_robust = TRUE, summarize metaVAR model using robust (sandwich) confidence intervals. |
| mplus          | Logical. If mplus = TRUE, summarize the DSEM model.                                                      |
| ncores         | Positive integer. Number of cores to use.                                                                |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SumFitMetaVARNormal      *Summary: Normal Theory (FitMetaVAR)*

---

### Description

Summary: Normal Theory (FitMetaVAR)

### Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

### Arguments

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| ncores        | Positive integer. Number of cores to use.                                                 |

### Details

This function is executed via the Sum function.

### Value

The output is saved as an external file in output\_folder.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

SumFitMetaVARRobust      *Summary: Robust (FitMetaVAR)*

---

### Description

Summary: Robust (FitMetaVAR)

### Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```



**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| ncores        | Positive integer. Number of cores to use.                                                 |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                           |
|-------------|---------------------------|
| SumFitMplus | <i>Summary (FitMplus)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMplus)

**Usage**

SumFitMplus(taskid, reps, output\_folder, overwrite, integrity, ncores)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| ncores        | Positive integer. Number of cores to use.                                                 |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                       |
|-------------|-----------------------|
| SumFitNaive | <i>Summary: Naive</i> |
|-------------|-----------------------|

---

**Description**

Summary: Naive

**Usage**

SumFitNaive(taskid, reps, output\_folder, overwrite, integrity, ncores)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| ncores        | Positive integer. Number of cores to use.                                                 |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

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